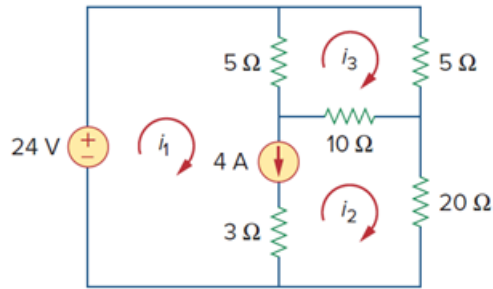


Problem

Use mesh analysis to determine i_1 , i_2 , and i_3 in Fig. 3.25.

Figure 3.25:



Step-by-step solution

Step 1 of 6

Refer to Figure 3.25 in the textbook.

The mesh 1 and 2 form a super mesh (dotted mesh) since they have an independent current source in common as shown in Figure 1.

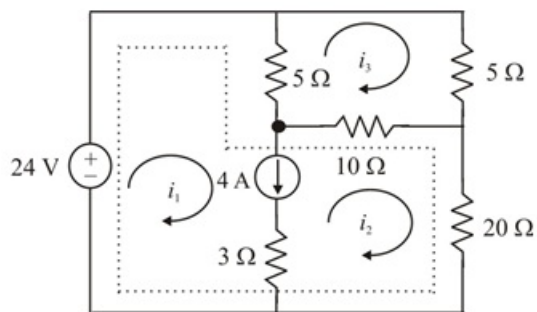


Figure 1

Step 2 of 6

Write the relation between the currents i_1 and i_2 .

$$i_1 - i_2 = 4$$

$$i_1 = 4 + i_2 \dots\dots (1)$$

Therefore, the expression for the current i_1 is, $4 + i_2$.

Step 3 of 6

Apply Kirchhoff's voltage law to the super mesh.

$$5(i_1 - i_3) + 10(i_2 - i_3) + 20i_2 - 24 = 0$$

$$5i_1 - 5i_3 + 10i_2 - 10i_3 + 20i_2 - 24 = 0$$

$$5i_1 + 30i_2 - 15i_3 - 24 = 0$$

Substitute $4 + i_2$ for i_1 .

$$5(4 + i_2) + 30i_2 - 15i_3 - 24 = 0$$

$$20 + 5i_2 + 30i_2 - 15i_3 - 24 = 0$$

$$35i_2 - 15i_3 - 4 = 0$$

$$35i_2 = 15i_3 + 4$$

$$i_2 = \frac{15i_3 + 4}{35} \dots\dots (2)$$

Step 4 of 6

Apply Kirchhoff's voltage law to the mesh 3.

$$5i_3 + 10(i_3 - i_2) + 5(i_3 - i_1) = 0$$

$$5i_3 + 10i_3 - 10i_2 + 5i_3 - 5i_1 = 0$$

$$20i_3 - 10i_2 - 5i_1 = 0$$

Substitute $4 + i_2$ for i_1 .

$$20i_3 - 10i_2 - 5(4 + i_2) = 0$$

$$20i_3 - 10i_2 - 20 - 5i_2 = 0$$

$$20i_3 - 15i_2 = 20$$

Substitute $\frac{15i_3 + 4}{35}$ for i_2 .

$$20i_3 - 15\left(\frac{15i_3 + 4}{35}\right) = 20$$

$$20i_3 - 6.43i_3 - 1.714 = 20$$

$$13.57i_3 = 21.714$$

$$i_3 = 1.6 \text{ A}$$

Therefore, the value of the current i_3 is **1.6 A**.

Step 5 of 6

Recall equation (2).

$$i_2 = \frac{15i_3 + 4}{35}$$

Substitute 1.6 A for i_3 .

$$i_2 = \frac{15(1.6) + 4}{35}$$

$$= \frac{28}{35}$$

$$= 0.8 \text{ A}$$

Therefore, the value of the current i_2 is **0.8 A**.

Step 6 of 6

Recall equation (1).

$$i_1 = 4 + i_2$$

Substitute 0.8 A for i_2 .

$$\begin{aligned} i_1 &= 4 + 0.8 \\ &= 4.8 \text{ A} \end{aligned}$$

Therefore, the value of the current i_1 is 4.8 A.